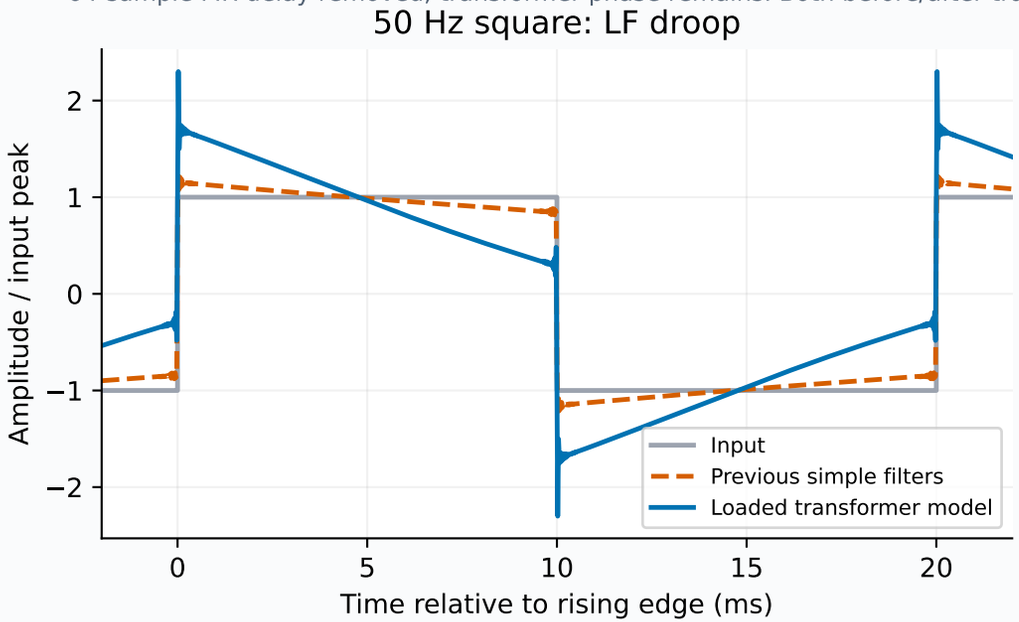
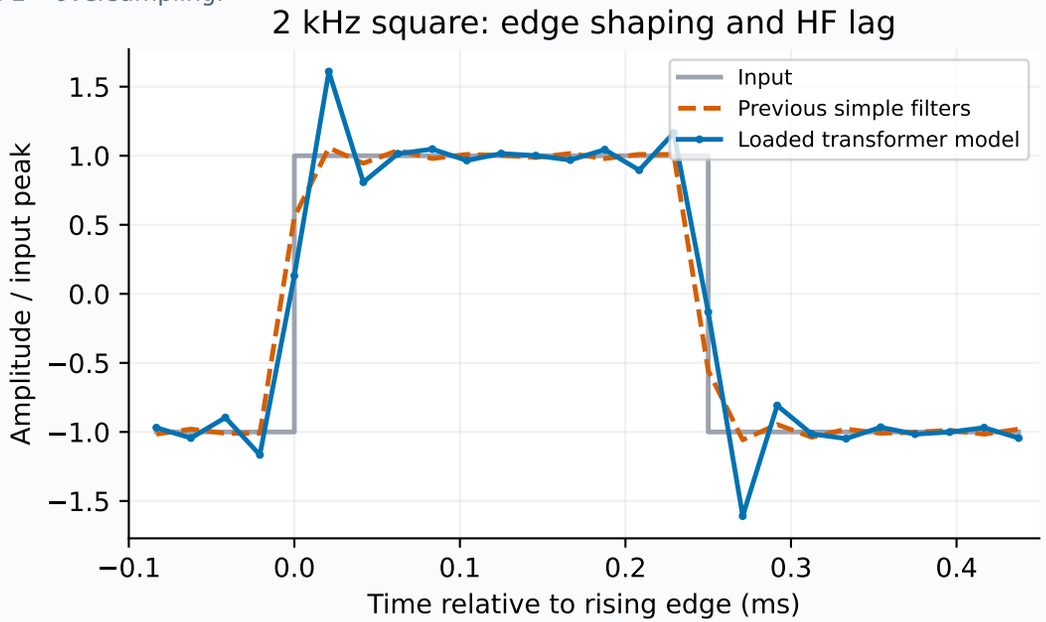
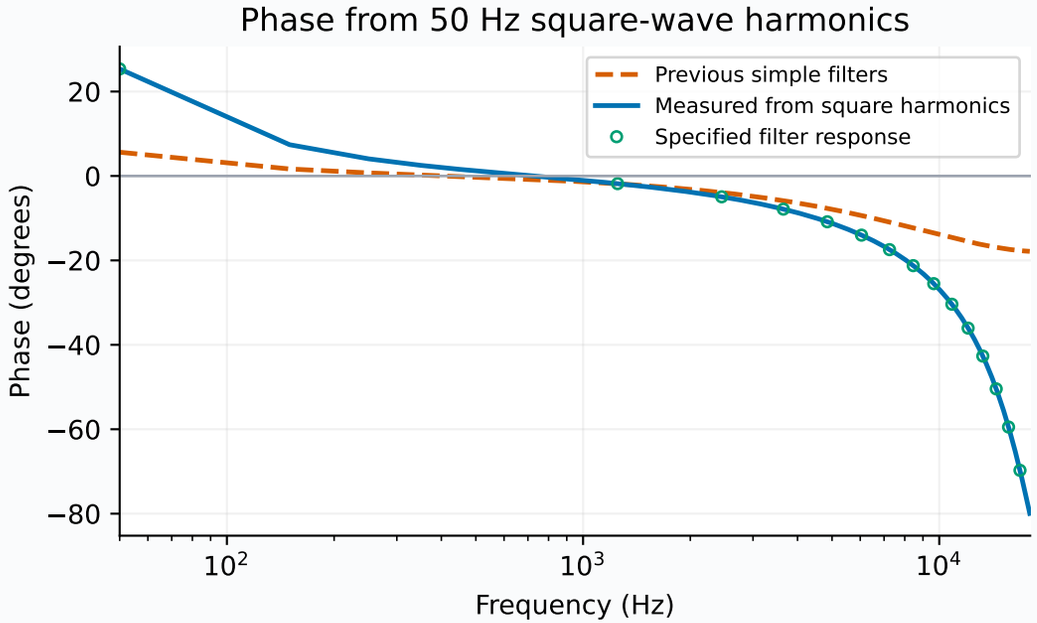
Transformer phase / square-wave verification

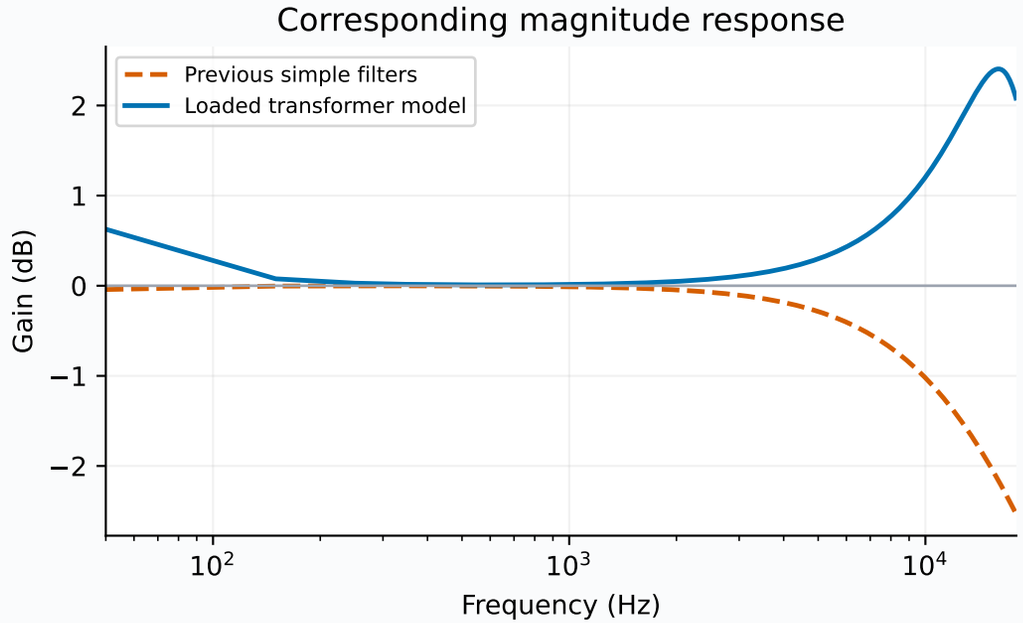
Actual DSP • 48 kHz host / 2× transformer • −60 dBFS peak • compression 1:1 • drive 0 dB

64-sample FIR delay removed; transformer phase remains. Both before/after traces use 2× oversampling.









LF model: 20 Hz, Q 1.0 • HF model: 19 kHz, Q 1.2 • maximum phase error vs design: 0.0170°

Behavioral loaded-transformer approximation; no specific hardware calibration. Low-level squares isolate phase from saturation.